

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE PHYSICS

Paper 2

Monday 14 November 2022 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the Physics Equations Sheet (enclosed).

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information

- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

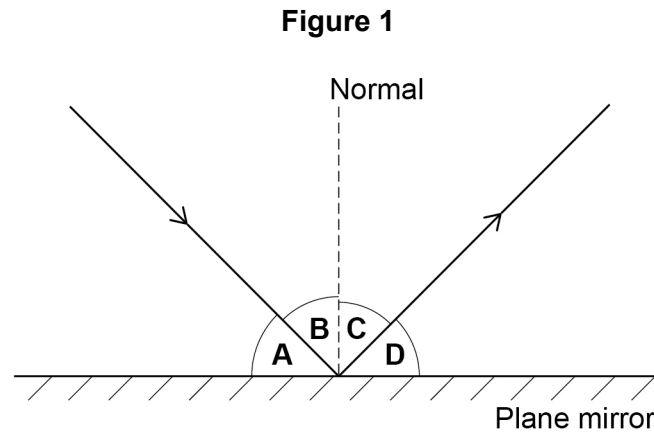


Answer **all** questions in the spaces provided.

0 1

A student investigated the reflection of light in a plane mirror.

Figure 1 shows a ray of light reflected by a plane mirror.



0 1 . 1

Which angle in **Figure 1** represents the angle of reflection?

[1 mark]

Tick (✓) **one** box.

A

☐

B

☐

C

☐

D

☐

0 1 . 2

What is the angle between the normal and the plane mirror?

[1 mark]

Tick (✓) **one** box.

0°

☐

45°

☐

90°

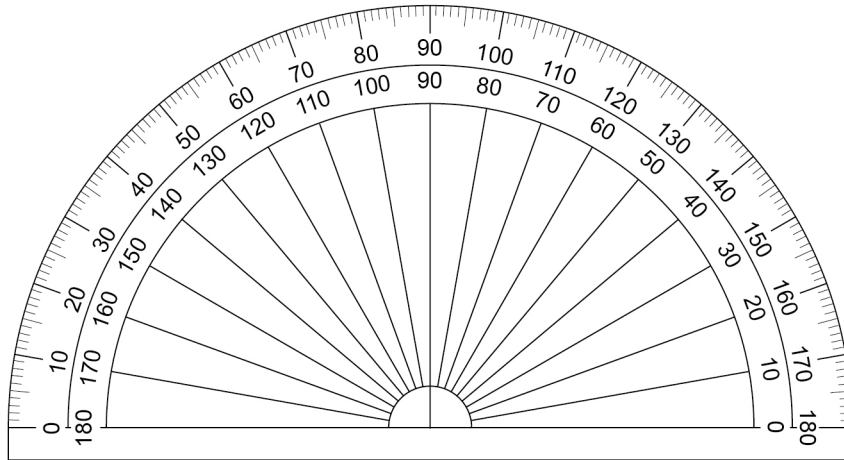
☐

180°

☐


Figure 2 shows the protractor the student used to measure the angles.

Figure 2



0 1 . 3 What is the resolution of the protractor in **Figure 2**?

[1 mark]

Tick (✓) **one** box.

0.1°

☐

1°

☐

10°

☐

90°

☐

0 1 . 4 What is the maximum angle that the protractor in **Figure 2** can measure?

[1 mark]

Question 1 continues on the next page

Turn over ►



Table 1 shows the results.

Table 1

Angle of incidence in degrees	Angle of reflection in degrees
10	10
20	21
30	29
40	40

0 1 . 5 Give a conclusion from the data in **Table 1**.

[1 mark]

0 1 . 6 Which of the following would improve the student's investigation?

[1 mark]

Tick (✓) **one** box.

Use a different coloured ray of light.

☐

Use a greater range of angles of incidence.

☐

Use a mirror made of thicker glass.

☐

Use a wider ray of light.

☐


0 1 . 7

Figure 3 shows an ambulance and the image of the ambulance in a mirror.

Figure 3



Ambulance

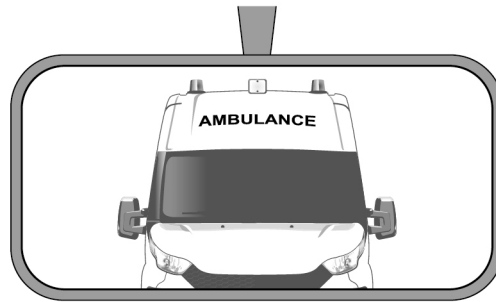


Image of the ambulance in a mirror

Why does the word 'ambulance' look different when viewed in the mirror?

[1 mark]

Tick (✓) **one** box.

The image in the mirror is laterally inverted.

☐

The image in the mirror is upright.

☐

The image in the mirror is virtual.

☐

7

Turn over for the next question

Turn over ►

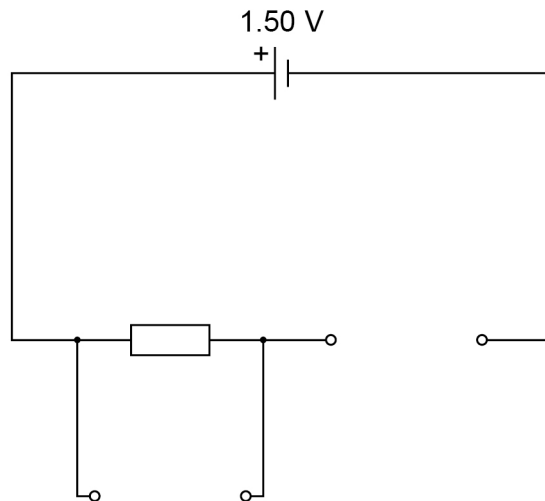


0 2

A student investigated how the potential difference across a resistor affects the current in the resistor.

Figure 4 shows an incomplete circuit diagram of the circuit the student used.

Figure 4



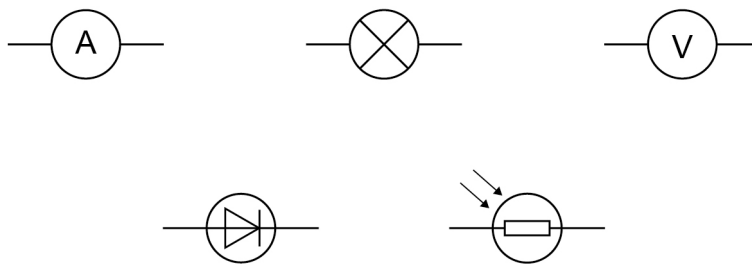
0 2 . 1

Complete **Figure 4** to show how to measure the potential difference across the resistor and the current in the resistor.

Use circuit symbols from **Figure 5**.

[2 marks]

Figure 5



0 2 . 2

The student varied the potential difference by adding more 1.50 V cells in series to the circuit.

What was the potential difference across the resistor when the student used 8 cells?

[1 mark]

_____ V

Potential difference = _____ V



0 2 . 3

When the student used a single 1.50 V cell, the current in the resistor was 0.075 A.

Calculate the resistance of the resistor.

Use the equation:

$$\text{resistance} = \frac{\text{potential difference}}{\text{current}}$$

[2 marks]

Resistance = _____ Ω

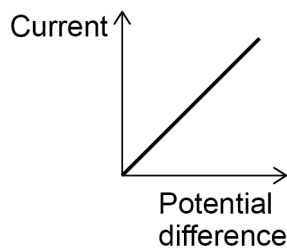
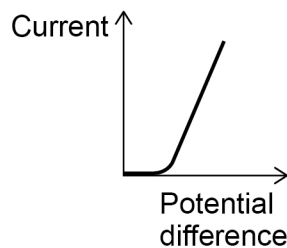
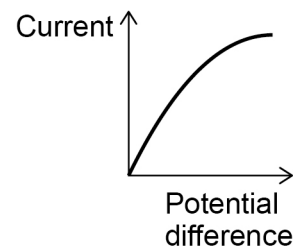
0 2 . 4

The resistor stayed at a constant temperature.

Which of the following is a graph for a resistor at constant temperature?

[1 mark]

Tick (✓) **one** box.


☐

☐

☐

0 2 . 5

What is **one** advantage of including the switch in the circuit in **Figure 4**?

[1 mark]

Tick (✓) **one** box.

It increases the current in the circuit.

☐

It increases the energy transferred from the cell.

☐

It increases the lifetime of the cell.

☐

Question 2 continues on the next page

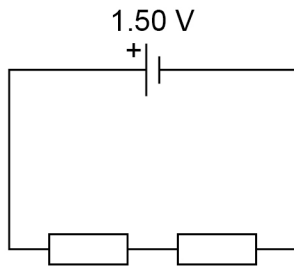
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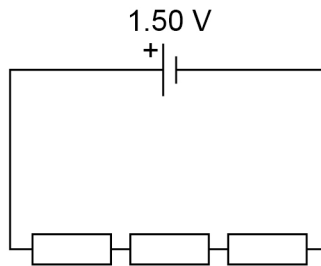
Figure 6 shows three different circuits.

All the resistors in **Figure 6** are identical.

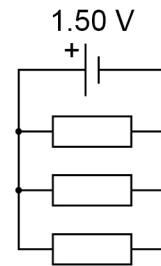
Figure 6



A



B



C

0 2 . 6 Which circuit has the greatest total resistance?

[1 mark]

Tick (✓) **one** box.

A

☐

B

☐

C

☐

0 2 . 7 Which circuit has the greatest current through the cell?

[1 mark]

Tick (✓) **one** box.

A

☐

B

☐

C

☐

0 2 . 8 What is the potential difference across each resistor in each circuit in **Figure 6**?

Complete the sentences.

Choose answers from the box.

[3 marks]

0.25 V	0.50 V	0.75 V	1.00 V	1.25 V	1.50 V
---------------	---------------	---------------	---------------	---------------	---------------

The potential difference across each resistor in circuit **A** is _____.

The potential difference across each resistor in circuit **B** is _____.

The potential difference across each resistor in circuit **C** is _____.

12

Turn over for the next question

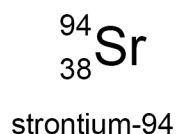
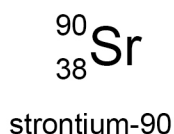
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0 3

There are different isotopes of strontium (Sr).

Two of the isotopes of strontium can be represented as:



0 3 . 1

An atom of strontium-90 contains 38 electrons, 52 neutrons and 38 protons.

Give the number of electrons, neutrons and protons in an atom of strontium-94

[2 marks]

Number of electrons _____

Number of neutrons _____

Number of protons _____

0 3 . 2

Strontium-94 is a radioactive isotope that emits beta radiation.

What is beta radiation?

[1 mark]

Tick (✓) **one** box.

A high-frequency electromagnetic wave.

☐

High-speed electrons from the nucleus.

☐

Particles consisting of two protons and two neutrons.

☐


0 3 . 3

When an atom of strontium-94 decays, it becomes an atom of yttrium-94

What happens to the proton number when a nucleus of strontium-94 decays?

[1 mark]

Tick (✓) **one** box.

The proton number decreases by one.

☐

The proton number stays the same.

☐

The proton number increases by one.

☐**0 3 . 4**

Strontium-94 has a half-life of 75 seconds.

Calculate the time taken for the count rate from a sample of strontium-94 to reduce to $\frac{1}{16}$ of its original value.

[3 marks]

Time taken = _____ s

Question 3 continues on the next page

Turn over ►

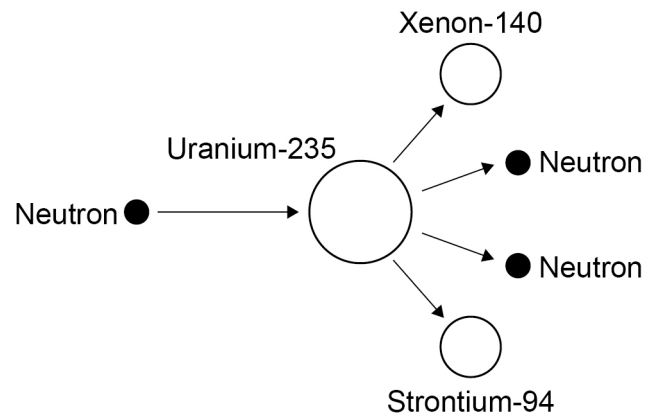


0 3 . 5

Strontium-94 is produced during the process of nuclear fission.

Figure 7 shows the process of nuclear fission.

Figure 7



Describe the process of nuclear fission shown in **Figure 7**.

[3 marks]

10



0 4

Ultrasound is a longitudinal wave.

0 4 . 1

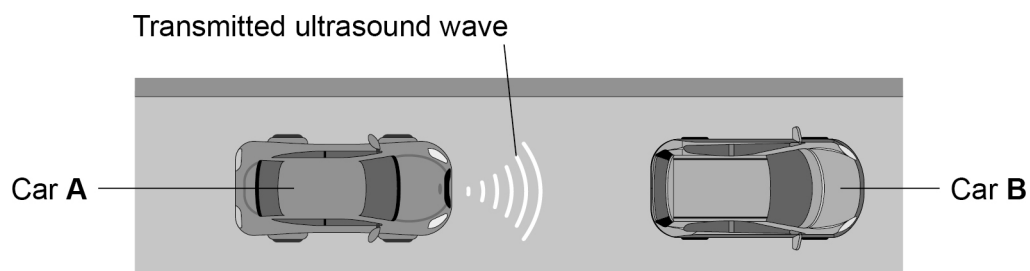
What is meant by a 'longitudinal wave'?

[2 marks]

0 4 . 2

Figure 8 shows ultrasound being used to measure the distance between two cars.

Figure 8



Explain how ultrasound can be used to measure the distance between two cars.

[3 marks]

Question 4 continues on the next page

Turn over ►

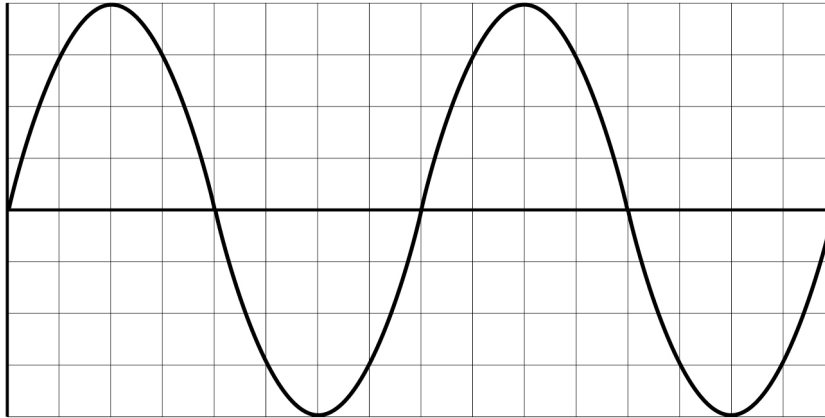


0 4 . 3

A microphone was connected to an oscilloscope.

Figure 9 shows the oscilloscope trace of the sound wave detected by the microphone.

Figure 9



Each horizontal division on the oscilloscope display represents a time of $50 \mu\text{s}$.

Determine if the trace in **Figure 9** represents an ultrasound wave.

Use the equation:

$$\text{frequency} = \frac{1}{\text{period}}$$

[4 marks]



0 5

Electricity is distributed from power stations along high-voltage transmission cables.

0 5 . 1

Explain the advantage of distributing electricity along high-voltage transmission cables.

[3 marks]

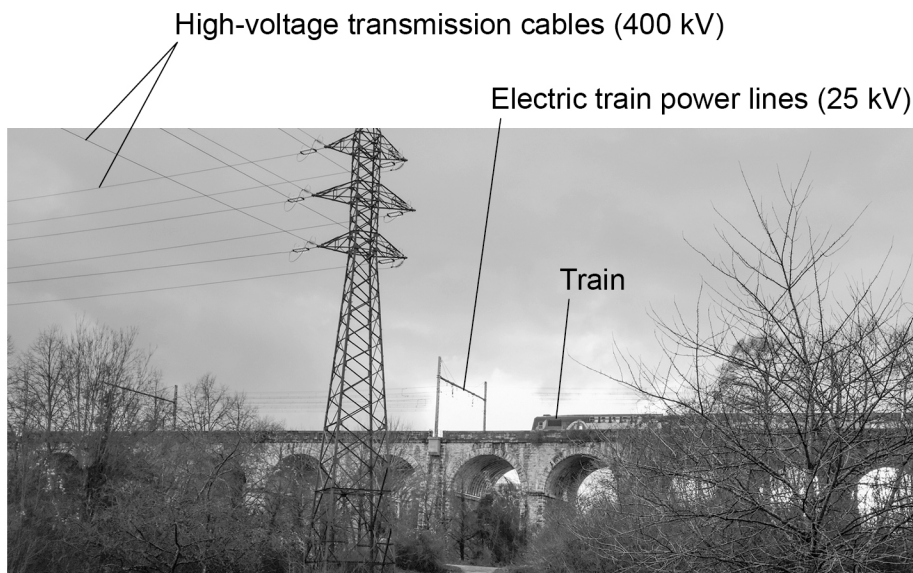
Question 5 continues on the next page

Turn over ►

A step-down transformer reduces the voltage for electric train power lines.

Figure 10 shows high-voltage transmission cables, electric train power lines, and an electric train.

Figure 10



0 5 . 2 When the train accelerates, the power of the train is 7.2 MW.

Determine the current provided to the train by the electric train power lines when the train accelerates.

Use information from **Figure 10**.

Use the Physics Equations Sheet.

[3 marks]

Current = _____ A



0 5 . 3

The current in the electric train power lines in **Figure 10** is 48 A when the train is travelling at a constant speed.

Determine the current the high-voltage transmission cables provide to the step-down transformer.

Use information from **Figure 10**.

Assume the step-down transformer is a basic transformer that is 100% efficient.

Use the Physics Equations Sheet.

[3 marks]

Current = _____ A

9

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0	6
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Figure 11 shows a person using a resistance band to exercise.

The resistance band is made from an elastic material.

Figure 11



The greater the force applied to the resistance band, the more the band extends.

Question 6 continues on the next page



Plan an investigation to determine if the force applied is directly proportional to the extension for the resistance band.

[6 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

A scientist did a similar investigation using a piece of wire.

Table 2 shows the results.

Table 2

Force in newtons	Extension in mm			
	Test 1	Test 2	Test 3	Mean
100	0.64	0.66	0.65	0.65
200	1.56	1.50	1.50	1.52
300	2.10	2.10	2.10	2.10
400	2.94	X	2.94	2.95
500	3.65	3.67	3.63	3.65
600	4.37	4.37	4.40	4.38
700	5.03	5.07	5.05	5.05

0 6 . 2

Explain why the scientist could **not** use a metre rule to measure the extension of the wire.

[2 marks]

0 6 . 3

Determine value **X** in **Table 2**.

[2 marks]

X = _____ mm



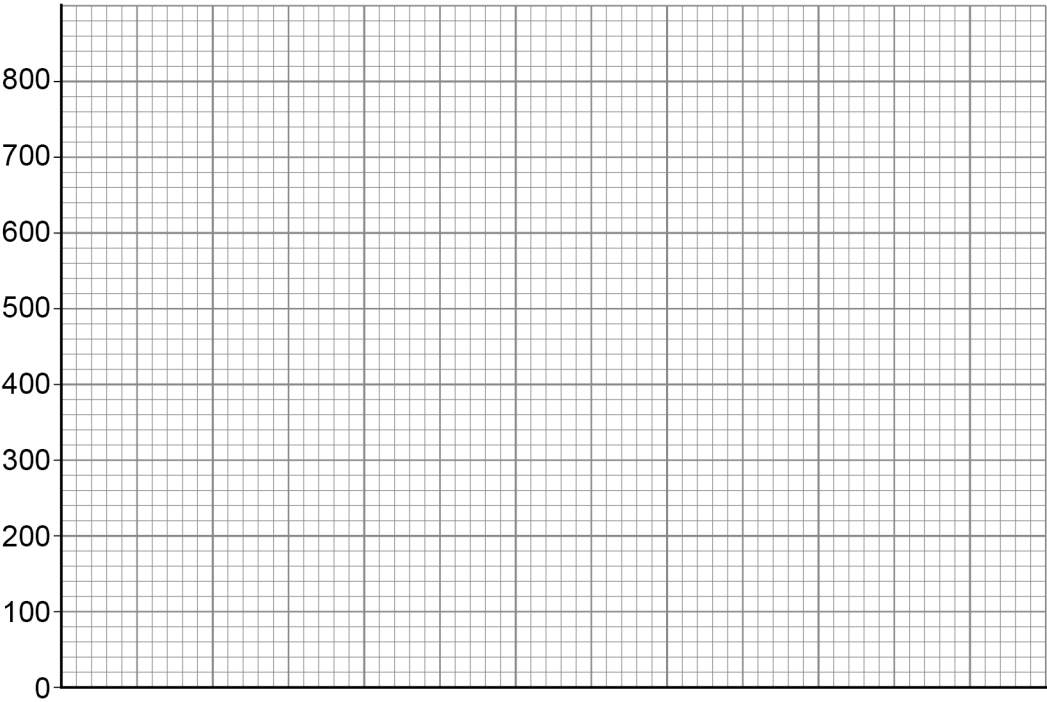
Table 2 is repeated below.

Table 2

Force in newtons	Extension in mm			
	Test 1	Test 2	Test 3	Mean
100	0.64	0.66	0.65	0.65
200	1.56	1.50	1.50	1.52
300	2.10	2.10	2.10	2.10
400	2.94	X	2.94	2.95
500	3.65	3.67	3.63	3.65
600	4.37	4.37	4.40	4.38
700	5.03	5.07	5.05	5.05

Figure 12 is an incomplete graph of the results.

Figure 12



0 6 . 4 Complete **Figure 12**.

You should:

- label both axes
- use a suitable scale for the x -axis
- plot the data from **Table 2**
- draw a line of best fit.

[5 marks]

0 6 . 5 The relationship between force (F) and extension (e) for the wire is:

$$F = k \times e$$

Determine the value of k using the gradient of your line of best fit on **Figure 12**.

Give your answer in N/m.

[3 marks]

$k =$ _____ N/m

18

Turn over for the next question

Turn over ►



0 7

In 2021 a small battery-powered helicopter performed the first ever powered flight on the planet Mars.

The mass of the helicopter was 1.8 kg.

0 7 . 1

The helicopter took off from the surface of Mars.

The helicopter then remained stationary above the surface for several seconds.

gravitational field strength on Mars = 3.7 N/kg

Calculate the upward force on the helicopter when it was stationary.

Use the Physics Equations Sheet.

[2 marks]

Upward force = _____ N



0 7 . 2

The helicopter then accelerated horizontally to its maximum speed.

At maximum speed, the kinetic energy of the helicopter was 8.1 J.

Determine the maximum speed of the helicopter.

Use the Physics Equations Sheet.

[3 marks]

Maximum speed = _____ m/s

0 7 . 3

The helicopter had a small mass.

A helicopter with a small mass can fly further than a helicopter with a large mass, when powered by the same battery.

Explain why.

Use ideas about energy in your answer.

[4 marks]

Question 7 continues on the next page

Turn over ►

Mars and Mercury both orbit the Sun.

Mars orbits at a greater distance from the Sun than Mercury.

0 7 . 4

Give the reason why the orbital velocity of Mars is less than the orbital velocity of Mercury.

[1 mark]

0 7 . 5

Table 3 gives data about the orbits of Mercury and Mars.

The orbital period is the time it takes a planet to make one complete orbit of the Sun.

Table 3

Planet	Distance from the Sun in Gm	Orbital velocity in km/s	Orbital period in days
Mercury	58	47	88
Mars	228	24	690

The orbital velocity of Mercury is approximately twice the orbital velocity of Mars.

Explain why the orbital period of Mars is approximately eight times the orbital period of Mercury.

Use data from **Table 3** and the following equation:

$$\text{circumference of a circle} = 2\pi r$$

[3 marks]



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0 8

Figure 13 shows an electric kettle with a metal case.

The kettle contains a heating element which is used to heat water.

Figure 13



0 8

1

The electric kettle has some safety features:

- earth wire
- fuse
- circuit breaker.

Explain each of these safety features.

[6 marks]



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